Notes: Allen, Morris, and Shin (RFS, 2006)

Beauty Contests and Iterated Expectations in Asset Markets

Contents

1 Big picture

- **Question.** Can Keynes's "beauty contest" metaphor be embedded in a fully rational asset-pricing model, and if so, what exactly do higher-order beliefs do to equilibrium prices?
- **Setting.** The paper studies a dynamic noisy rational-expectations market with one risky asset, short-lived risk-averse traders, private signals, public information, and noisy supply. The asset is traded repeatedly before its final liquidation value is realized.
- **Core challenge.** In a representative-investor martingale world, today's price folds all future payoffs back into the present through the law of iterated expectations. With differential information, however, the relevant pricing object is not one trader's conditional expectation but *average expectations*, and average expectations generally do *not* satisfy the law of iterated expectations.
- **Main conceptual insight.** Once the law of iterated expectations fails for average beliefs, prices can depend on *average expectation of average expectations*, and therefore on higher-order beliefs. This is the rational benchmark that makes Keynes's intuition operational.
- **Main result 1.** The mean path of prices deviates systematically from the market's average expectation of fundamentals. Prices can therefore contain a "higher-order wedge" relative to consensus fundamental value.
- **Main result 2.** Prices respond too sluggishly to changes in the fundamental. Even though they are forward-looking, they display inertia or drift. This gives the model an outward resemblance to short-run momentum.
- **Main result 3.** Public information receives too much weight in prices relative to private information. The reason is that public information helps forecast *others'* expectations and therefore aggregate demand.
- **Why this matters.** The paper shows that higher-order beliefs do not require irrationality, behavioral agents, or explicit coordination motives. They arise in a standard competitive setting once information is heterogeneous and traders care about resale prices.
- **Economic takeaway.** Rational traders with short horizons end up worrying about what others think others think. That creates a systematic gap between prices and fundamentals and makes prices especially sensitive to public signals.

2 Data and measurement (key objects)

2.1 No empirical dataset: this is a theory paper

- The paper does not estimate the model structurally and does not introduce a new empirical dataset.
- Its empirical role is instead to explain several familiar phenomena in asset markets: excess sensitivity to public information, sluggish adjustment of prices to news, and bubble-like departures of prices from fundamental consensus.
- The object being "measured" inside the model is not a sample moment from data, but the relationship between prices, higher-order expectations, and the final liquidation value.

2.2 Key random objects in the background example

- **Fundamental.** A scalar payoff variable $\theta \sim N(y, 1/\alpha)$, where y is the public prior mean and α is prior precision.
- **Private signal.** Each agent observes

$$x_i = \theta + \varepsilon_i, \qquad \varepsilon_i \sim N(0, 1/\beta), \tag{1}$$

  where β is signal precision.

- **Individual expectation.**

$$E_i(\theta) = \frac{\alpha y + \beta x_i}{\alpha + \beta}. \tag{2}$$

- **Average expectation.**

$$E(\theta) = \frac{\alpha y + \beta \theta}{\alpha + \beta}. \tag{3}$$

- **Public expectation.** The paper denotes by $E^*(\theta)$ the expectation conditional on public information only.

Interpretation. At the individual level, the private and public signals are blended in the usual Bayesian way. At the *average* level, however, idiosyncratic private-signal noise washes out, so public information becomes relatively more important.

2.3 Higher-order average expectations

Iterating the average-belief operator gives

$$E^k(\theta) = \left(1 - \left(\frac{\beta}{\alpha + \beta}\right)^k\right) y + \left(\frac{\beta}{\alpha + \beta}\right)^k \theta. \tag{4}$$

- This sequence converges to y as $k \to \infty$.
- So the higher the order of average belief, the more weight gets put on public information and the less weight on the true fundamental.

Interpretation. This is the paper's cleanest "measurement" result. It says that repeated averaging of beliefs systematically pushes beliefs toward the public signal.

2.4 Key objects in the dynamic asset-pricing model

- **Liquidation value.** The risky asset pays θ at date $T + 1$.
- **Prices.** The asset is traded at dates $1, 2, \ldots, T$, with market price p_t at date t.
- **Supply noise.** Net supply at date t is a normal random variable s_t with mean zero and precision γ_t.
- **Private signal at date t.**

$$x_{it} = \theta + \varepsilon_{it}, \qquad \varepsilon_{it} \sim N(0, 1/\beta). \tag{5}$$

- **Trader information.** A trader born at date t observes the prior mean y, the whole history of prices $p_1, \ldots, p_t$, and a private signal x_{it}.
- **Risk tolerance.** Traders have CARA utility $u(c) = -e^{-c/r}$, so r is the key parameter governing demand aggressiveness.

2.5 Time structure and why short horizons matter

- Traders live for two periods only: they buy when young and unwind the position when old.
- Because they consume before the asset's final liquidation, they care about the next period's resale price, not just the terminal payoff.
- This is the mechanism that makes higher-order beliefs enter equilibrium pricing in a natural way.

Interpretation. If traders only cared about the ultimate dividend and could ignore interim prices, the beauty-contest force would be much weaker. Short horizons turn beliefs about future prices into a first-order object.

3 Models and empirical specifications (all equations + interpretation)

3.1 Background: failure of the law of iterated expectations for average beliefs

The law of iterated expectations holds for individual and public expectations:

$$E_{it}(E_{i,t+1}(\theta)) = E_{it}(\theta), \qquad E_t^*(E_{t+1}^*(\theta)) = E_t^*(\theta). \tag{6}$$

But it generally fails for average beliefs:

$$\bar{E}_t(\bar{E}_{t+1}(\theta)) \neq \bar{E}_t(\theta). \tag{7}$$

In the static Gaussian example,

$$\bar{E}(\bar{E}(\theta)) = \left(1 - \left(\frac{\beta}{\alpha+\beta}\right)^2\right) y + \left(\frac{\beta}{\alpha+\beta}\right)^2 \theta, \tag{8}$$

which is biased toward the public signal y relative to the one-shot average expectation $\bar{E}(\theta)$.

Interpretation. This is the paper's basic engine. Once average expectations do not collapse under iteration, higher-order beliefs survive in equilibrium asset prices.

3.2 Asset market environment

- Time is discrete: $t \in \{1, 2, \ldots, T, T+1\}$.
- There is one risky asset, traded at dates 1 through T and liquidated at $T+1$.
- The fundamental θ is fixed from date 1 onward.

- At each trading date there is one young generation buying and one old generation selling.

Interpretation. The overlapping-generations structure is not an incidental technical trick. It is exactly what turns the problem into one of pricing *resale value* under heterogeneous beliefs.

3.3 Trader demand and market clearing

At the final trading date T, trader i's demand is

$$x_{iT} = \frac{r}{\mathrm{Var}_{iT}(\theta)}\Big(E_{iT}(\theta) - p_T\Big). \tag{9}$$

Aggregating across traders gives market demand

$$x_T = \frac{r}{\mathrm{Var}_T(\theta)}\Big(E_T(\theta) - p_T\Big), \tag{10}$$

so market clearing implies

$$p_T = E_T(\theta) - \frac{\mathrm{Var}_T(\theta)}{r} s_T. \tag{11}$$

At an earlier date $t < T$, a young trader cares about next period's price p_{t+1}, so demand is

$$x_{it} = \frac{r}{\mathrm{Var}_{it}(p_{t+1})}\Big(E_{it}(p_{t+1}) - p_t\Big), \tag{12}$$

which aggregates to

$$p_t = E_t(p_{t+1}) - \frac{\mathrm{Var}_t(p_{t+1})}{r} s_t. \tag{13}$$

Interpretation. Prices are not just discounted expectations of terminal dividends. They are expectations of *future prices*, and those future prices themselves embed higher-order expectations.

3.4 Recursive price formula and the beauty-contest structure

By recursively substituting the one-step price equation backward from date T, the paper obtains

$$p_t = E_t E_{t+1} \cdots E_T(\theta) - \frac{\mathrm{Var}_t(p_{t+1})}{r} s_t. \tag{14}$$

Interpretation.

- The first term says today's price is the date-t average expectation of the date-$(t+1)$ average expectation, and so on, of the final liquidation value.
- That is the fully rational beauty-contest formula: price depends on beliefs about beliefs about beliefs.
- The second term is the usual noisy-supply discount coming from risk aversion.

3.5 Signal extraction from prices

The history of prices can be transformed into a sequence of noisy public signals $\{\xi_1, \dots, \xi_t\}$ about θ. Given those transformed signals, trader i's posterior can be written as

$$E_{it}(\theta) = \frac{\beta x_{it} + \alpha y + \sum_{j=1}^{t} \rho_j \xi_j}{\beta + \alpha + \sum_{j=1}^{t} \rho_j}, \tag{15}$$

where ρ_j denotes the precision of the jth price signal.

Interpretation. Prices become public statistics about fundamentals. But because those prices are themselves determined by others' beliefs, the price history carries a higher-order component that a purely Bayesian forecast of θ alone would miss.

3.6 Matrix representation and the nonhomogeneous Markov chain

The paper stacks the public prior, the transformed price signals, and the fundamental into a state vector

$$z = (y, \xi_1, \dots, \xi_T, \theta)'. \tag{16}$$

Then average expectations can be represented by a date-specific stochastic matrix B_t:

$$E_t z = B_t z, \tag{17}$$

so iterated average expectations are generated by repeated multiplication of the B_t matrices:

$$E_t E_{t+1} \cdots E_T z = B_t^* z, \tag{18}$$

where B_t^* is the product of the relevant transition matrices.

Interpretation.

- The public prior y is an absorbing state.
- The price signals ξ_j are temporary absorbing states: initially they trap probability mass, but later they become transient.
- The more one iterates average expectations, the more probability mass drifts toward the public-information state.

This Markov-chain representation is the paper's main technical contribution.

3.7 Linear price representation

The equilibrium price can be written as a linear combination of the prior, the fundamental, and the history of supply shocks:

$$p_t = \lambda_t y + \kappa_t \theta - \sum_{j=1}^{t} \phi_{tj} s_j, \qquad \lambda_t + \kappa_t = 1. \tag{19}$$

A similar expression holds for the average expectation of fundamentals:

$$E_t(\theta) = \Lambda_t y + (1 - \Lambda_t)\theta - \sum_{j=1}^{t} \psi_{tj} s_j. \tag{20}$$

Interpretation. The entire problem boils down to how much weight prices and beliefs put on the public prior y versus the true payoff θ. The paper's propositions compare those weights.

3.8 Main propositions

Proposition 1. For all $t < T$,

$$E_s\big|p_t - \theta\big| > E_s\big|E_t(\theta) - \theta\big|, \tag{21}$$

with equality only at date T.

Interpretation. The price path is *farther from fundamentals* than the market's average expectation of fundamentals. That is the paper's bubble-like result.

Proposition 2. There exist weights $\{\lambda_t\}$ with

$$0 < \lambda_T < \cdots < \lambda_2 < \lambda_1 < 1 \tag{22}$$

and

$$E_s(p_t) = \lambda_t y + (1 - \lambda_t)\theta. \tag{23}$$

Interpretation. Prices exhibit inertia. After the fundamental is realized, the initial response is too small, and the mean price path drifts gradually toward θ.

Proposition 3. Define the naive price path

$$q_t = \left(1 - \left(\frac{\beta}{\alpha + \beta}\right)^{T-t+1}\right) y + \left(\frac{\beta}{\alpha + \beta}\right)^{T-t+1} \theta. \tag{24}$$

Then as traders become arbitrarily risk averse ($r \to 0$),

$$E_s(p_t) \to q_t. \tag{25}$$

Interpretation. When traders are very risk averse, prices reveal almost nothing. The equilibrium then collapses toward the naive higher-order-expectations path that ignores information extracted from prices.

3.9 Long-lived traders and robustness

The paper also studies a two-period long-lived-trader version. In the limit of very noisy supply, expected first- and second-period prices coincide, so the first-period trader behaves as if the second trading opportunity effectively does not exist because it is hedged away by future demand.

Interpretation.

- The short-lived assumption strengthens the beauty-contest force, but the qualitative mechanism does not disappear in longer-horizon environments.
- What matters is not literally overlapping generations per se. What matters is that some traders care about interim prices before all future payoffs are realized.

4 Identification and credibility (what makes the mechanism credible)

4.1 What drives the results

- **Differential information.** Without heterogeneous information, there is no gap between individual and average iterated expectations.
- **Noisy supply.** Prices are informative but not fully revealing because supply shocks prevent perfect inference from price alone.
- **Short horizons.** Traders care about resale prices, which makes higher-order expectations directly relevant for demand.
- **Risk aversion.** Finite risk tolerance keeps demand slopes finite and prevents arbitrage from instantly forcing prices to equal fundamentals.

4.2 Why public information gets overweighted

- Public information is common to everyone, so it is especially useful for forecasting *average opinion.*
- Private information is informative about θ, but its idiosyncratic noise gets washed out in the aggregate.
- The public signal therefore matters twice: directly for forecasting fundamentals and indirectly for forecasting others' demand.

Interpretation. The paper does not assume an explicit coordination motive. The coordination force is implicit in equilibrium pricing because everyone knows the public signal enters everyone else's demand.

4.3 Why this is a rational asset-pricing result rather than a behavioral one

- The paper keeps preferences standard and beliefs Bayesian.
- There are no irrational traders, no ad hoc sentiment shocks, and no explicit noise-trader demand.
- The beauty-contest phenomenon comes entirely from equilibrium under asymmetric information.

Interpretation. This is why the paper is important. It rationalizes a famous Keynesian intuition without leaving the standard noisy rational-expectations tradition.

4.4 Relation to the martingale property

- In representative-investor models, the law of iterated expectations and the martingale property let future outcomes be folded cleanly into today's price.
- In this paper, the average-expectations operator fails to satisfy the law of iterated expectations.

- If average expectations are the relevant pricing operator, there need not exist an equivalent martingale measure of the familiar representative-investor type.

Interpretation. The paper's point is not just that the martingale property fails. It is that it fails in a *systematic* direction: toward public information and away from private information.

4.5 What makes the results robust

- As private signals become more precise, prices get closer to fundamentals.
- As traders become less risk averse, demand becomes more aggressive and prices become more revealing.
- With long-lived traders, the qualitative force survives as long as interim consumption or liquidity needs make resale prices matter.

4.6 Main limitations

- The paper is deliberately stylized: one risky asset, Gaussian signals, CARA utility, and linear equilibrium.
- The strongest results come from short-lived traders; quantitatively, their force could be weaker in richer long-horizon environments.
- The model captures one aspect of bubbles — systematic departure from common-knowledge fundamentals — but not dramatic boom-bust dynamics by itself.
- There is no empirical estimation, so the paper is a mechanism paper rather than a measurement paper.

4.7 Relation to the literature

- Relative to the classic Grossman–Hellwig noisy rational-expectations tradition, the paper highlights higher-order beliefs rather than just information aggregation.
- Relative to Bacchetta and Van Wincoop, it provides a clean dynamic framework and the higher-order wedge interpretation.
- Relative to behavioral bubble models, it shows that rational traders with differential information can generate beauty-contest outcomes on their own.
- Relative to macro models of higher-order expectations, it provides a tractable fully solved asset-pricing application.

5 Tables and figures

5.1 Figure 1 and Figure 2: timing and information structure

Read:

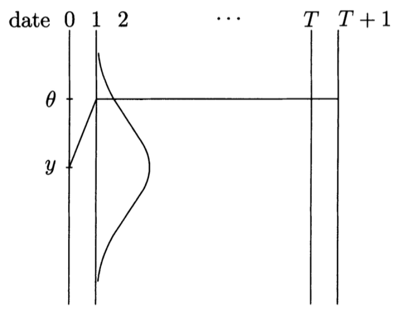

Figure 1
Path of fundamental value

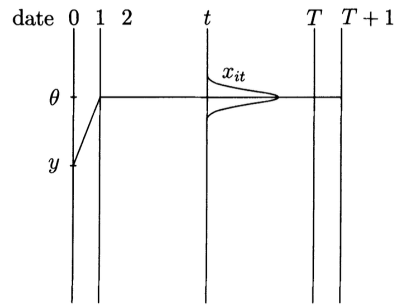

Figure 2
Private information

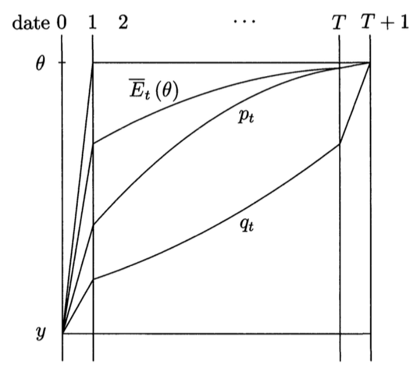

Figure 3
Mean of time paths of p_t and $\bar{E}_t(\theta)$

- Figure 1 fixes the timing of the fundamental. The liquidation value θ is realized before trading begins and then stays constant until the terminal liquidation date.
- Figure 2 shows what each newborn trader knows: the full history of prices and one private signal at birth.
- These two figures explain why the paper is not about learning from changing fundamentals. The dynamics come from how information about a fixed fundamental gets aggregated through prices.

5.2 The background example: Equations (1)–(3)

Read:

- The background Gaussian example is the simplest demonstration that repeated average expectations drift toward the public prior y.
- The associated naive price path shows immediate public-signal bias even before the full asset-pricing model is solved.
- This example is important because it reveals the mechanism in one line: higher-order beliefs put progressively more weight on public information.

5.3 Figure 3 and Propositions 1–2: bubble-like deviation and inertia

Read:

- Figure 3 is the central visual in the paper. It plots three objects: the mean price path $E_s(p_t)$, the mean path of average expectations $E_s(E_t(\theta))$, and the naive higher-order path q_t.
- The price path lies below the average-expectations path early on, so prices underreact relative even to market consensus.
- Proposition 1 says the mean price path is farther from the true fundamental than the mean path of average expectations.
- Proposition 2 says that the weights on the public prior decline only gradually, so the price path drifts toward fundamentals over time.

Bottom line. The equilibrium price does not jump to the consensus fundamental value. It moves only gradually, which is why the model looks like momentum despite being fully rational.

5.4 Proposition 3: the limiting case of very noisy prices

Read:

- Proposition 3 says that as risk tolerance goes to zero, the mean price path approaches the naive path q_t.
- In that limit, prices stop being useful signals because traders are too timid to trade aggressively on information.
- This proposition clarifies the role of price informativeness: the more informative prices are, the more the actual price path can depart from the naive higher-order benchmark.

5.5 Proposition 7 and the discussion section

Read:

- The long-lived-trader comparison shows that the first-round price in the short-run model is more biased toward the public signal than in the long-run model.
- The discussion section makes the broader point that some traders consuming early is enough to recreate the same qualitative mechanism.
- The paper also connects its results to the nonexistence of an equivalent martingale measure under genuine asymmetric information and to the literature on rational bubbles.

6 Short summary

- **Method.** The paper builds a dynamic noisy rational-expectations asset-pricing model with heterogeneous information, short-lived risk-averse traders, and noisy prices.
- **Main conceptual result.** Average expectations do not satisfy the law of iterated expectations, so prices depend on higher-order beliefs in a systematic way.

- **Main equilibrium result.** Prices overweight public information, deviate from consensus fundamental value, and adjust too sluggishly to news.
- **Key quantitative statements.** The mean price path is farther from fundamentals than the mean path of average expectations, it drifts gradually toward fundamentals, and it converges toward the naive higher-order path when prices become uninformative.
- **Economic lesson.** Keynesian beauty-contest reasoning is not just metaphorical. It appears naturally in a fully rational competitive asset market once traders have differential information and care about short-run resale prices.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that higher-order beliefs matter in a fully rational asset-pricing model.
- It also establishes that the relevant failure is not irrationality but the failure of the law of iterated expectations for *average* beliefs under asymmetric information.
- As a result, equilibrium prices can differ systematically from both fundamentals and the market's average expectation of fundamentals.

7.2 Economic intuition

- Traders care about tomorrow's price because they sell before the asset liquidates.
- Tomorrow's price depends on others' beliefs, so today's demand depends on beliefs about others' beliefs.
- Public information becomes disproportionately valuable because it helps forecast average opinion and aggregate demand.
- That is why prices become overly public-signal-driven and why they adjust sluggishly to a change in the true payoff.

7.3 Takeaway

This paper is important because it converts Keynes's beauty-contest metaphor into a clean rational-expectations theorem. Once prices are expectations of future prices and information is dispersed, higher-order beliefs become a first-order determinant of asset values. The result is a systematic higher-order wedge: prices can move away from common-knowledge fundamentals, react too much to public news, and drift only gradually back toward the true liquidation value.